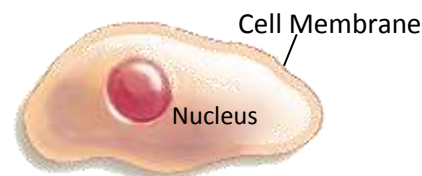


Diffusion across a Selectively Permeable Membrane^{1 2}

Each cell is surrounded by a **selectively permeable cell membrane** which regulates what gets into and out of the cell. A selectively permeable membrane allows some types of molecules and ions to diffuse across the membrane and prevents other types of molecules and ions from crossing the membrane. For example, oxygen can cross the selectively permeable cell membrane, but large molecules like proteins and DNA cannot cross the cell membrane.



1. Why is it useful for each cell to be surrounded by a selectively permeable cell membrane?

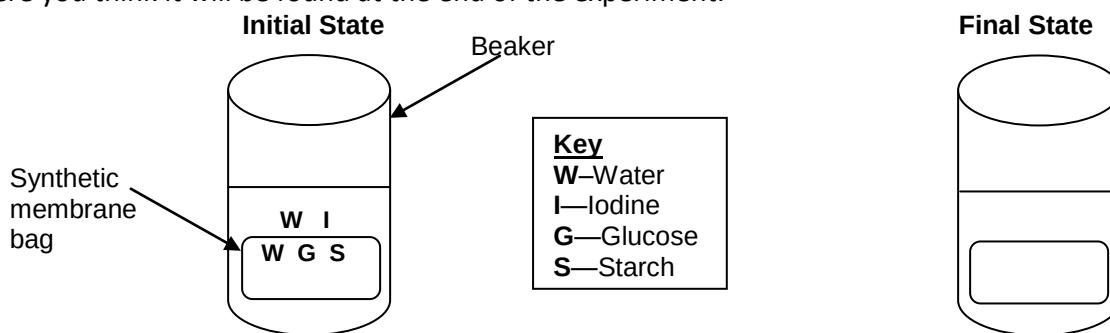
Today you will investigate a synthetic selectively permeable membrane which provides a simplified model of the cell membrane. Specifically, you will test the hypothesis that smaller molecules and ions can cross this synthetic selectively permeable membrane, but larger molecules cannot.

2. Which of the following molecules do you think will diffuse across the selectively permeable membrane?

Molecule or Ion (Molecular Formula)	Will it cross the membrane?	Why or why not?
Iodine (I_3^-)		
Water (H_2O)		
Glucose ($C_6H_{12}O_6$)		
Starch (polysaccharide made up of many molecules of glucose)		

To test your predictions, you will put solutions of starch and glucose in a bag made of the synthetic membrane and put the bag in a beaker of iodine solution. You will allow time for the substances to diffuse across the membrane and then test which of the substances have crossed the membrane.

3. The Initial State diagram shows the locations of each type of molecule or ion at the beginning of the experiment. Based on your answers to question 2, predict where each type of molecule or ion will be found after diffusion. In the Final State diagram, write the letter for each type of molecule or ion in the places where you think it will be found at the end of the experiment.



¹ A bold number on the left indicates a question you should answer and an > arrow indicates something you should do.

² By Drs. Jennifer Doherty and Ingrid Waldron, Department of Biology, University of Pennsylvania, © 2015. Teachers are encouraged to copy this Student Handout for classroom use. A Word file, which can be used to prepare a modified version if desired, an alternative version of the Student Handout with a more complete introduction to osmosis, and Teacher Preparation Notes with instructional suggestions and background information are available at <http://serendip.brynmawr.edu/exchange/waldron/diffusion>.

To test whether iodine or starch have crossed the synthetic membrane, you will look for a change in color. A solution of iodine is tan and a solution of starch is clear or milky white; when iodine and starch are together in the same solution, the solution is purple, dark blue or black.

To test whether glucose has crossed the synthetic membrane, you will use a glucose test strip to test for glucose in the solution in the beaker.

If water can cross the synthetic membrane, water could diffuse into the bag or out of the bag. Your teacher will let you know how you will test for change in volume of water in the bag.

4. For each substance, indicate how you will know whether it crossed the synthetic membrane. What observation will be different, depending on whether or not each substance crossed the membrane?

Substance	Expected Observation	
	If this substance crossed the membrane	If this substance did <u>not</u> cross the membrane
Iodine		
Starch		
Glucose		
Water		

Procedure:

- Obtain a piece of pre-soaked dialysis tubing and two pieces of string. Fold the bottom of the piece of tubing 1 cm up and use a piece of string to tie the folded part tightly to create a bag.
- To open the other end of the tube, rub the end between your fingers until the edges separate. Use pipettes to add 4 mL of glucose solution and 4 mL of starch solution to the tube.
- Next, fold 1 cm of tubing at the top of the bag and tie it off tightly. Check to make sure there are no leaks. If scissors are available, trim the strings short.
- Rinse the bag and strings thoroughly in fresh water. Dry the bag and strings thoroughly.
- In the table below record your initial observation of the measure you are using to evaluate the movement of water.
- Add 200 mL of distilled water to a 250 mL beaker. Add about 0.8 mL iodine to the water in the beaker.
Caution: Be careful to avoid staining your clothes or fingers with the iodine.
- Put the bag in the beaker.

5. Record your observations of the colors of the solution in the bag and the solution in the beaker in the "Initial State" row in the table below.

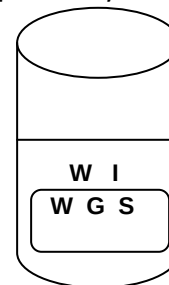
	In the bag		In the beaker		Measure to evaluate movement of water
	Color	Glucose?	Color	Glucose?	
Initial State		Yes		No	
Final State		Yes			

You will need to wait 20-30 minutes for substances to move across the selectively permeable membrane.

While you're waiting, read the following paragraph and answer question 6.

Osmosis is the movement of water across a selectively permeable membrane. Osmosis results in the net movement of water from a solution with a lower concentration of solutes to a solution with a higher concentration of solutes. In your experiment, the concentration of solutes is higher in the starch and glucose solution in the bag than in the iodine solution in the beaker.

6. Which way would you expect more water to diffuse – into the bag or out of the bag? Explain why.



➤ After 20-30 minutes, record your observations in the “Final State” row of the table on the bottom of page 2.

7. Complete this table.

Molecule or Ion (Molecular Formula)	Did this molecule or ion cross the membrane?	How do you know?
Iodine (I_3^-)		
Water (H_2O)		
Glucose ($C_6H_{12}O_6$)		
Starch (polysaccharide made up of many molecules of glucose)		

8. Based on your results, what characteristic appears to determine which molecules and ions can cross the synthetic selectively permeable membrane?

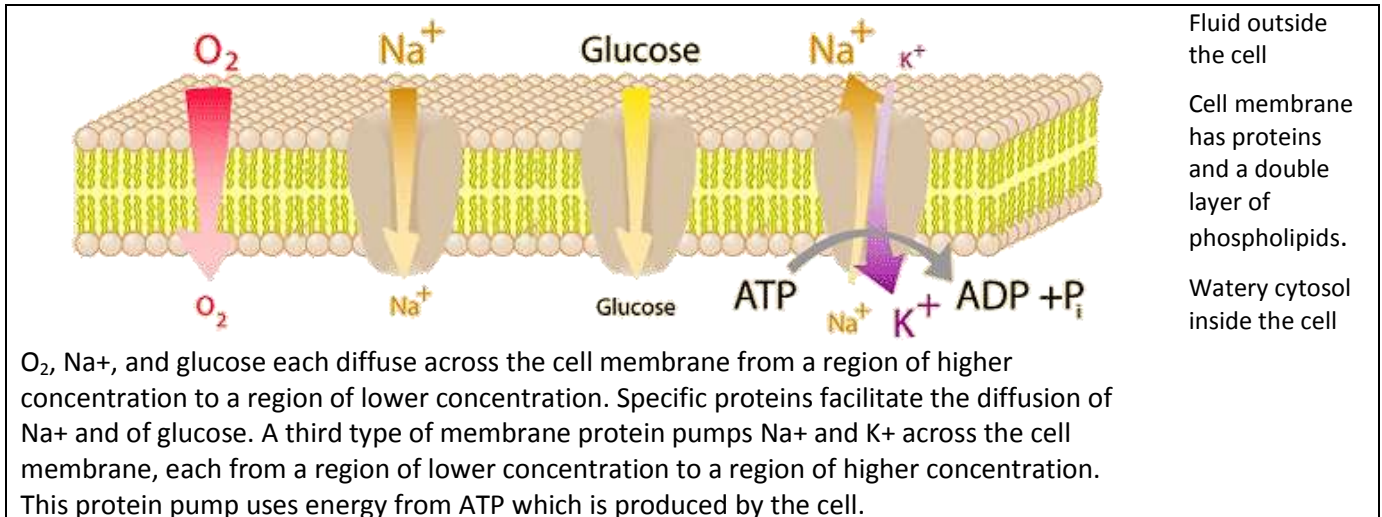
9a. Did any of your results differ from your predictions in questions 3 and 6? If so, what were the differences between your results and your predictions?

9b. What do you think is the explanation for any differences between your predictions and your observations?

A **model** is a simplified representation of a complex biological structure or process. A model focuses on a few key features in order to help us understand a biological structure.

Because a model is simpler than the biological structure it represents, a model does not show all the features of the actual biological structure. For example, the synthetic membrane in your experiment is a model of the selectively permeable cell membrane, but it shows a much simpler kind of selective permeability than an actual cell membrane has.

This diagram provides a different model of the cell membrane. It gives some idea of the complexity of the selective permeability of an actual cell membrane. The diagram shows that a cell membrane contains proteins that aid in the transport of biologically important ions and molecules across the cell membrane.



This activity has presented two models of the selectively permeable cell membrane: this diagram and the synthetic membrane in your experiment. Both models show that small molecules and ions can cross a selectively permeable membrane. However, there are some differences in other features of these models.

10. The fourth type of transport shown in the diagram is observed in living cells, but not in a synthetic membrane. Explain why this type of transport is possible in living cells, but not in a synthetic membrane.

11. Which feature of cell membranes was demonstrated by your experiment with the synthetic membrane, but is not shown in the diagram? (Hint: Think about starch.)